



ML FOR TEXT MINING

Section 2

What is Parsing?

The Definition: The process of analyzing a sequence of words to determine its grammatical structure.

The Goal: To help computers understand not just *what* words are used, but *how* they relate to one another.

The Analogy: If POS tagging is identifying the parts of a car (engine, wheel, bolt), parsing is understanding how they are assembled to make the car move.

Parsing Techniques

Shallow Parsing (Chunking)

Groups words into meaningful **chunks** (e.g., noun phrases, verb phrases), providing a higher-level structural view.

Constituency Parsing

Identifies the **hierarchical phrasal structure** of a sentence, showing how words group together into nested components (phrases).

Dependency Parsing

Identifies **grammatical relationships** between words, showing which words modify or depend on others in a sentence.

Shallow Parsing (Chunking)

Definition: Groups words into meaningful, non-overlapping "chunks" (e.g., noun phrases, verb phrases) without analyzing the internal structure or the relationship between the chunks.

How it works: It uses **Part-of-Speech (POS) tags** as patterns.

Example rule: NP → Det Adj Noun.

Key Insight: It is **Flat and Non-recursive**. A chunk cannot contain another chunk (no nesting).

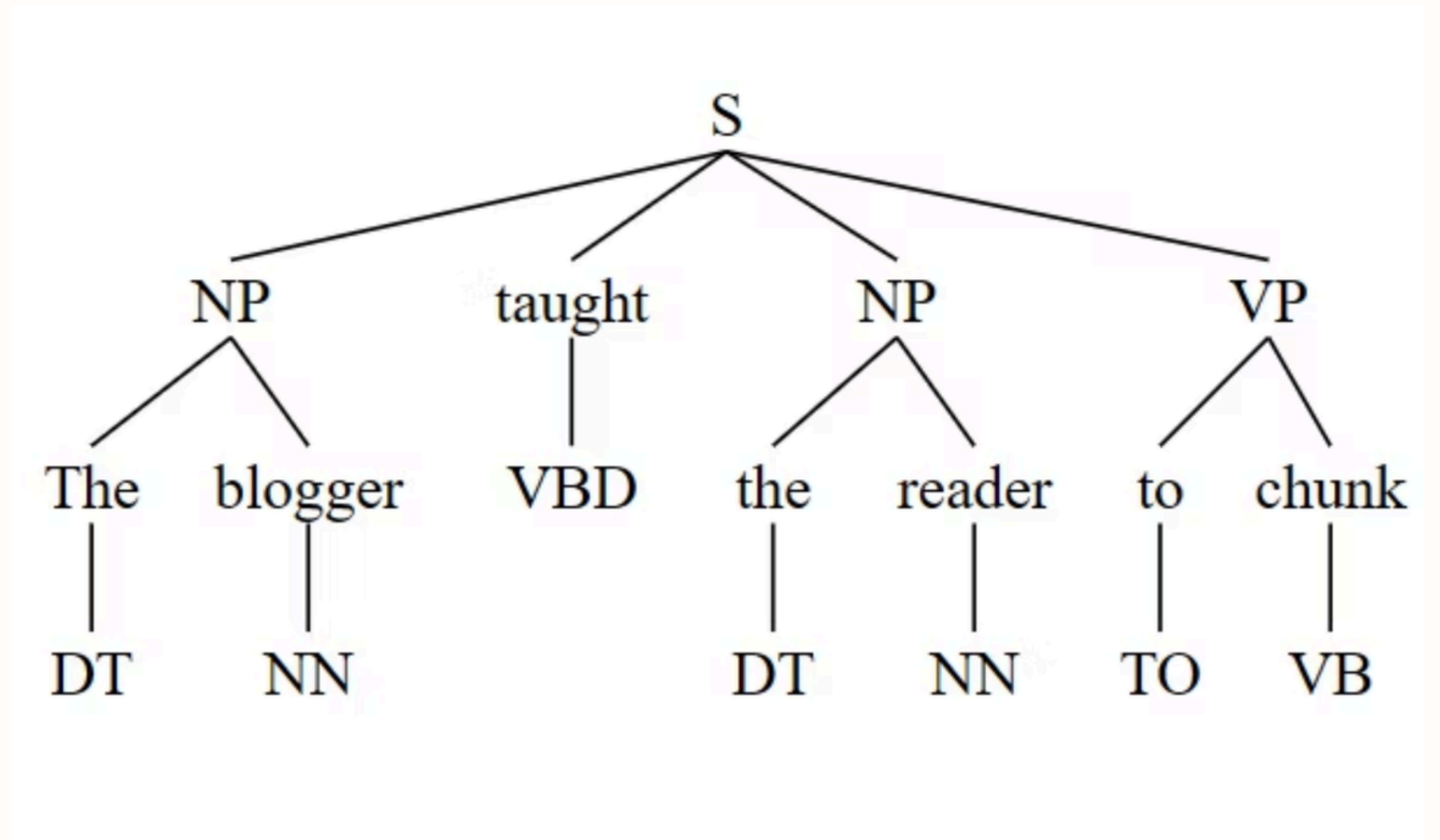
Shallow Parsing (Chunking)

The blogger taught the reader to chunk

[NP The blogger] (Noun Phrase)

[NP the reader] (Noun Phrase)

[VP to chunk] (Verb Phrase)



Constituency Parsing

Definition: Breaks a sentence down into a **hierarchy** of nested components (constituents). It views a sentence as a set of "boxes within boxes."

How it works: It uses **Context-Free Grammar (CFG)** rules to build a tree.

Example rules:

- $S \rightarrow NP + VP$
- $NP \rightarrow Det + N$
- $VP \rightarrow V + NP$

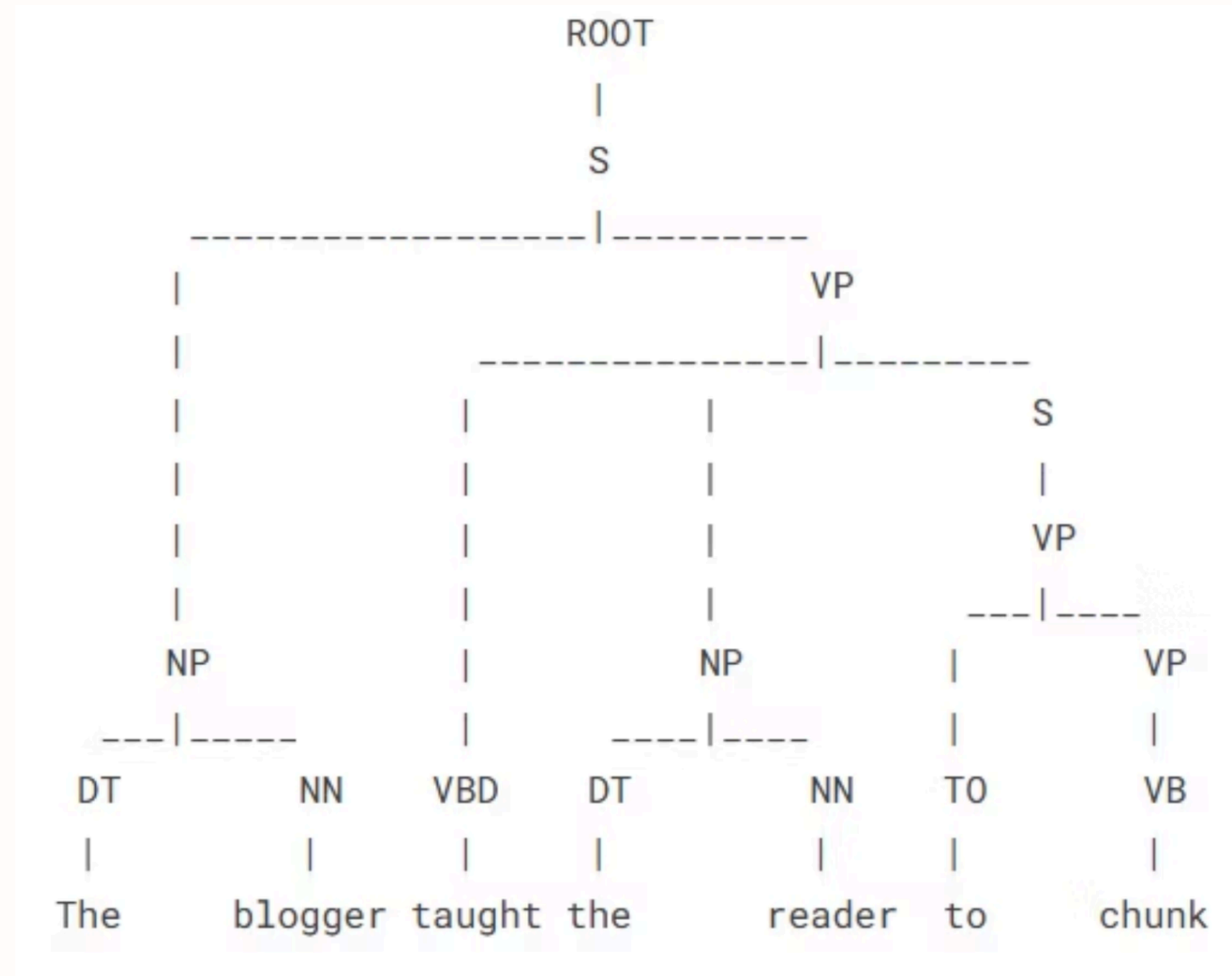
Key Insight: It is **recursive**. You can have a phrase inside a phrase. It explains the "Architecture" of language—how small units build into larger, complex thoughts.

Constituency Parsing

Example Analysis:

The blogger taught the reader to chunk

It would show that "**to chunk**" is a sub-phrase nested inside the larger Verb Phrase "**taught the reader to chunk**".



Dependency Parsing

Definition: Identifies direct **binary relationships** between words. It ignores "phrases" and focuses on the functional link between a **Head** (the governor) and its **Dependents** (the modifiers).

The Logic: Every word in a sentence serves a specific purpose for another word.

- The ROOT (usually the main verb) is the anchor of the entire sentence.
- Every other word is a Dependent that "reports" to exactly one Head.

How it works: Instead of boxes, it uses **Directed Arcs** (arrows) to map how words connect. If you remove a Head, its Dependents no longer have a grammatical function in the sentence.

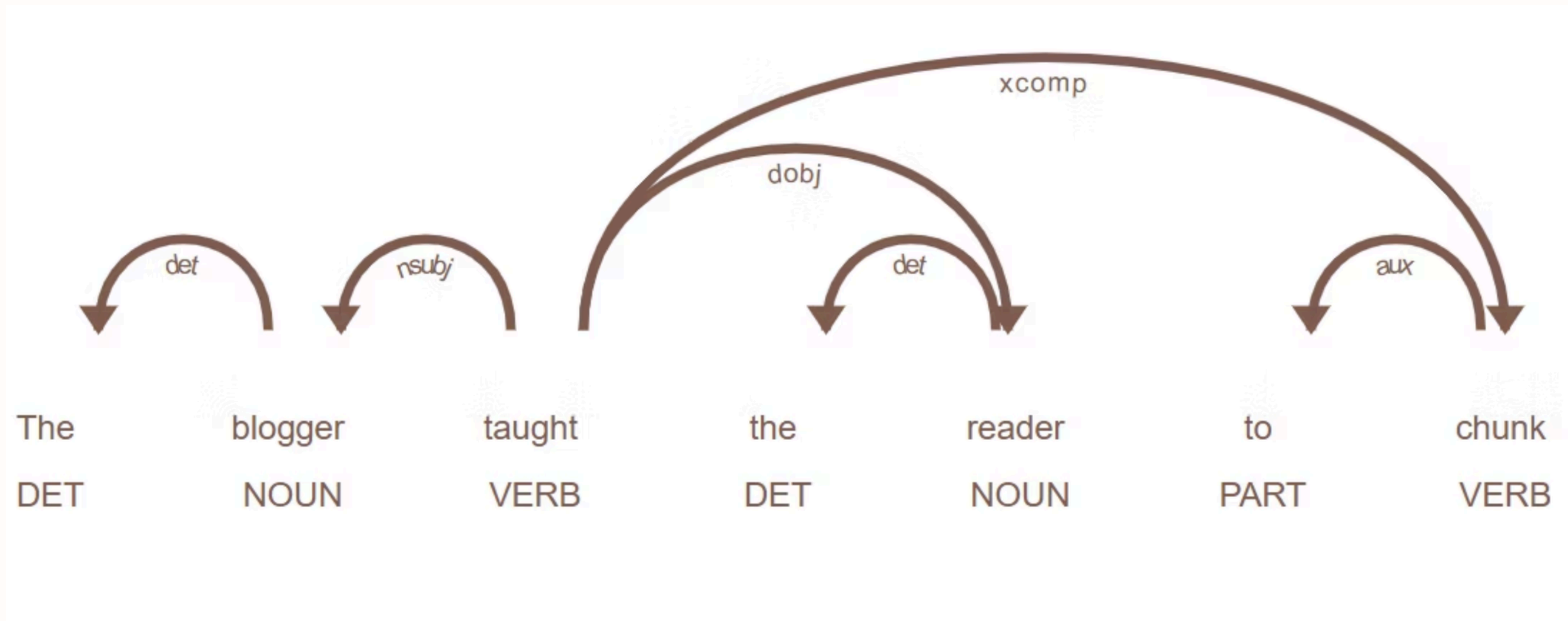


Key Insight: It focuses on **Semantics** (meaning) rather than just Syntax (arrangement).

Dependency Parsing

Example Analysis:

The blogger taught the reader to chunk



Note: The direction of the arcs (arrows) is almost always **from the Head to the Dependent**.

Summary Comparison

Feature	Shallow Parsing	Constituency Parsing	Dependency Parsing
Analogy	The Highlighter: Grouping neighbors.	The Blueprint: Building an architecture.	The Map: Linking roles.
Structure	Flat / Linear	Hierarchical / Nested	Relational / Directed
Core Question	"What are the basic units?"	"How are phrases nested?"	"Which word governs which?"
Output	[Chunks]	(Tree (Nodes))	Word → Word
Relationship	Proximity (POS patterns)	Membership (Grammar rules)	Governance (Functional roles)

Thank You

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